INCH-POUND

MIL-S-19500/408D
25 August 1992
SUPERSEDING
MIL-S-19500/408C
12 May 1980

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER
TYPES 2N3715 AND 2N3716 JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, high-power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

	P_T 1/ T_A = +25°C	P_T T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W max
2N3715	5.0	85.7	60	60	7.0	4.0	10	-65 to +200	1.17
2N3716	5.0	85.7	100	80	7.0	4.0	10	-65 to +200	1.17

1/ Derate linearly 28.57 mW/°C above T_A = +25°C.
2/ Derate linearly 0.857 W/°C above T_C = +100°C.

1.4 Primary electrical characteristics.

	h_{FE1} 1/	h_{FE4} 1/	$V_{BE(SAT)1}$ 1/	$V_{CE(SAT)1}$ 1/	C_{obo}	$\|h_{fe}\|$
	V_{CE} = 2.0 V dc I_C = 3.0 A dc	V_{CE} = 4.0 V dc I_C = 10 A dc	I_C = 5.0 A dc I_B = 0.5 A dc	I_C = 5.0 A dc I_B = 0.5 A dc	V_{CB} = 10 V dc I_E = 0 f = 1 MHz	V_{CE} = 10 V dc I_C = 0.5 A dc f = 1 MHz
			V dc	V dc	pF	
Min	30	5.0				4.0
Max	120		1.5	1.0	500	20

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Ft. Monmouth, NJ 07703-5302 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500.

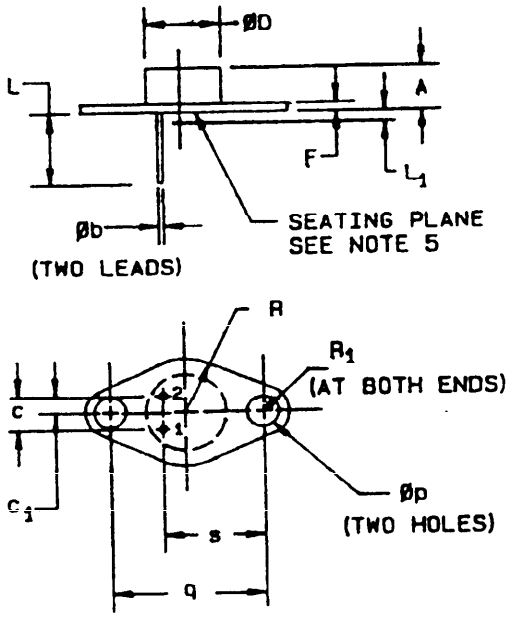
3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the country of origin marking may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.270	.380	6.86	9.65	
Øb	.038	.053	0.97	1.09	7
ØD		.875		22.22	
c	.420	.440	10.67	11.18	4
c_1	.205	.225	5.21	5.72	4
F	.060	.055	1.52	1.40	
L	.312	.500	7.92	12.70	
L_1		.050		1.27	
Øp	.151	.165	3.84	4.19	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal 1, base; terminal 2, emitter; case, collector.
4. These dimensions should be measured at points .050-.055 inch (1.27-1.40 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
5. The seating plane of the header shall be flat within .004 inch (0.10 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .006 inch (0.15 mm) concave to .006 inch (0.15 mm) convex overall.
6. Collector shall be electrically connected to the case.
7. Øb applies between L_1 and L. Diameter is uncontrolled in L_1.

FIGURE 1. Physical dimensions of transistor types 2N3715 and 2N3716.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal response (see 4.3.2)	Thermal response (see 4.3.2)
9	I_{CES1} and h_{FE2}	I_{CES1}
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater. Δh_{FE2} = ±15 percent of initial value.	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13	ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater; Δh_{FE2} = ±15 percent of initial value; subgroup 2 of table I herein.	ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater; Δh_{FE2} = ±20 percent of initial value; subgroup 2 of table I herein.

1/ This test shall be performed any time before screen 3.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5 ±12.5°C; V_{CE} = 25 ±5 V dc; T_A = ≤ 100°C.

4.3.2 Thermal response (ΔV_{BE}) measurements. The ΔV_{BE} measurements shall be performed in accordance with method 3131 of MIL-STD-750. The ΔV_{BE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor. The chosen ΔV_{BE} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen V_{BE} shall be plotted. The chosen V_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. One hundred percent safe operating area (SOA) testing may be performed in lieu of thermal response testing herein provided that the appropriate conditions of temperature, time current and voltage to achieve die attach integrity are submitted to the qualifying activity. The following parameter measurements shall apply:

a. I_M . . . . . 10 mA.

b. V_{CE} measurement voltage . . . . . 20 V (same as V_H).

c. I_H collector heating current . . . . . 4 A (minimum).

d. V_H collector-emitter heating voltage . . . . . 20 V (minimum).

e. t_H heating time . . . . . 100 ms.

f. t_{MD} measurement delay time . . . . . 50 µs to 80 µs.

g. t_{SW} sample window time . . . . . 10 µs (maximum).

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection (JANS). Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

a. Subgroup 1: Physical dimensions; method 2066; conditions: See figure 1.

b. Subgroup 2: Bond strength; method 2037; conditions: Test condition A; all internal wires shall be pulled separately.

c. Subgroup 4: Intermittent operation life; method 1037; conditions: V_{CB} = 30 V dc, P_T = 4 W at T_A = +25°C ±3°C, $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air on the device shall be permitted.

d. Subgroup 5:

A separate sample may be pulled for each test.

Accelerated steady-state operation life, method 1027, conditions: V_{CB} = 30 V dc, T_A = +125°C ±25°C for 96 hours; P_T = 4 W at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = 225°C. Marking legibility requirement shall not apply.

e. Subgroup 6: Thermal resistance, method 3131, conditions: See 4.5.2.

4.4.2 Group B inspection (JANTX and JANTXV). Group B inspection shall be conducted in accordance with conditions specified for the subgroup testing in table IVb (JANTX and JANTX) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

a. Subgroup 3:

Intermittent operation life, method 1037, conditions: $V_{CB} \geq 10$ V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off} \geq 1$ minute for 2,000 cycles. No heat sink or forced air cooling on the device shall be permitted.

Bond strength, method 2037, conditions: LTPD = 20.

b. Subgroup 5: Thermal resistance, method 3131, conditions: See 4.5.2.

c. Subgroup 6: High-temperature life (non-operating), method 1032, conditions: T_A = +200°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IV of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

a. Subgroup 1: Physical dimensions; method 2066; conditions: See figure 1.

b. Subgroup 2:

Thermal shock (glass strain), method 1056, conditions: Test condition B.

Terminal strength (tension), method 2036, conditions: Test condition A, weight = 10 lbs, application time = 15 seconds.

c. Subgroup 6: Intermittent operation life, method 1037, conditions: $V_{CB} \geq 10$ V dc; ΔT_J between cycles ≥ 100°C, $t_{on} = t_{off} \geq 1$ minute for 6,000 cycles. No heat sink or forced air cooling on the device shall be permitted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector-emitter breakdown voltage	3011	Bias cond. D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3715				60		V dc
2N3716				80		V dc
Collector-base cutoff current	3036	Bias cond. D;	I_{CBO}			
2N3715		V_{CB} = 80 V dc			10	µA dc
2N3716		V_{CB} = 100 V dc			10	µA dc
Emitter-base cutoff current	3061	Bias cond. D; V_{EB} = 7 V dc;	I_{EBO}		1.0	mA dc
Collector-emitter cutoff current	3041	Bias cond. A;	I_{CEX1}			
2N3715		V_{BE} = 1.5 V dc; V_{CE} = 80 V dc;			1.0	mA dc
2N3716		V_{CE} = 100 V dc			1.0	mA dc
Collector-emitter cutoff current	3041	Bias cond. C;	I_{CES1}			
2N3715		V_{CE} = 50 V dc			1.0	mA dc
2N3716		V_{CE} = 70 V dc			1.0	mA dc
Base-emitter saturated voltage	3066	Test cond. A; I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.5	V dc
Base-emitter saturated voltage	3066	Test cond. A; I_C = 10 A dc; I_B = 2 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$		3.0	V dc
Collector-emitter saturated voltage	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.0	V dc
Collector-emitter saturated voltage	3071	I_C = 10 A dc; I_B = 2 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		2.5	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE1}	50	150	
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE2}	30	120	
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE3}	10		
Forward-current transfer ratio	3076	V_{CE} = 4.0 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE4}	5		
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias cond. C;	I_{CES2}			
2N3715		V_{CE} = 50 V dc			5.0	mA dc
2N3716		V_{CE} = 70 V dc			5.0	mA dc
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE5}	12		
Subgroup 4						
Switching parameters						
Pulse delay time		See figure 2	t_d		0.2	µs
Pulse rise time		See figure 2	t_r		1.3	µs
Pulse storage time		See figure 2	t_s		1.2	µs
Pulse fall time		See figure 2	t_f		1.2	µs
t_{off}		$t_s + t_f$	t_{off}		2.0	µs

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 1 kHz	h_{fe}	30	300	
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 100 kHz - 1 MHz	$\|h_{fe}\|$	4.0	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; f = 1 MHz	C_{obo}		500	pf
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t ≥ 1 s; 1 cycle; (see figure 3)				
Test 1		V_{CE} = 15 V dc; I_C = 10 A dc				
Test 2		V_{CE} = 40 V dc; I_C = 3.75 A dc				
Test 3						
2N3715		V_{CE} = 55 V dc; I_C = 0.9 A dc				
2N3716		V_{CE} = 65 V dc; I_C = 0.9 A dc				
Safe operating area (clamped inductive)		T_A = +25°C; I_C = 10 A dc; V_{CC} = 15 V dc; (see figures 5 and 6)				
2N3715		Clamp voltage = 60 V dc				
2N3716		Clamp voltage = 80 V dc				
Safe operating area (unclamped inductive)	3053	Load cond. C (see fig. 4); T_A = +25°C; duty cycle ≤ 10 percent				
Test 1		t_p approx 1.0 ms (vary to obtain I_C); R_{BB1} = 5Ω; V_{BB1} = 8 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 5 A dc; L = 4 mH; Q ≥ 100 at 1 kHz; .05 ohms, 20 A dc; (Stanford Miller CK-20 or equivalent)				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Test 2		t_p approx 2.2 ms (vary to obtain I_C); R_{BB1} = 5Ω; V_{BB1} = 8 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 2.2 A dc; L = 20 mH; Q ≥ 6000 at 10 kHz; 0.22 ohms, 12.5 A dc; (Two Stancor C-2688 in series or equivalent)				
Safe operating area (clamped inductive) (destructive)		T_A = +25°C; I_C = 10 A dc; V_{CC} = 55 V dc; (see figures 5 and 6)				
2N3715		Clamp voltage = 60 V dc				
2N3716		Clamp voltage = 80 V dc				
Electrical measurements		See table II, steps 1 and 4.				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector-emitter cutoff current	3041	Bias condition C;	I_{CES1}			
	2N3715		V_{CE} = 50 V dc			10	µA dc
	2N3716		V_{CE} = 70 V dc			10	µA dc
2.	Collector-emitter (voltage saturated)	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		1.0	V dc
3.	Base-emitter voltage saturated	3066	Bias condition A; I_C = 5.0 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.5	V dc
4.	Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE2}	30	120	
5.	Collector-emitter cutoff current	3041	Bias condition C;	ΔI_{CES1} 4/	100% of initial value or 1 µA dc; whichever is greater.		
	2N3715		V_{CE} = 50 V dc				
	2N3716		V_{CE} = 70 V dc				
6.	Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	Δh_{FE2} 4/	±25% change from initial value		
7.	Collector-emitter (voltage saturated)	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$\Delta V_{CE(SAT)1}$ 4/	±50 mV dc change from previously measured value		
8.	Thermal response	3131	See 4.3.2.	ΔV_{BE}			

See footnotes at end of table.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements 1/ 2/ 3/ - Continued.

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, 3, and 4.

b. Subgroup 4, see table II herein, steps 1, 2, 3, 4, 7, and 8.

c. Subgroup 5, see table II, steps 1, 2, 3, 4, 5, 6, and 7.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, and 3.

b. Subgroup 3, see table II herein, steps 1, 3, 4, 5, and 6.

c. Subgroup 6, see table II herein, steps 1, 4, 5, and 6.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 2, 3, and 4 (JANS); and 1, 2, and 4 (JANTX and JANTXV).

b. Subgroup 3, see table II herein, steps 1, 2, 3, and 4 (JANS); and 1, 2, and 4 (JANTX and JANTXV).

c. Subgroup 6, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, and 8 (JANS); and 1, 2, 3, 4, 5, and 6 (JANTX and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be acceptable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply.

a. Collector current magnitude during power application shall be 1.0 A dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

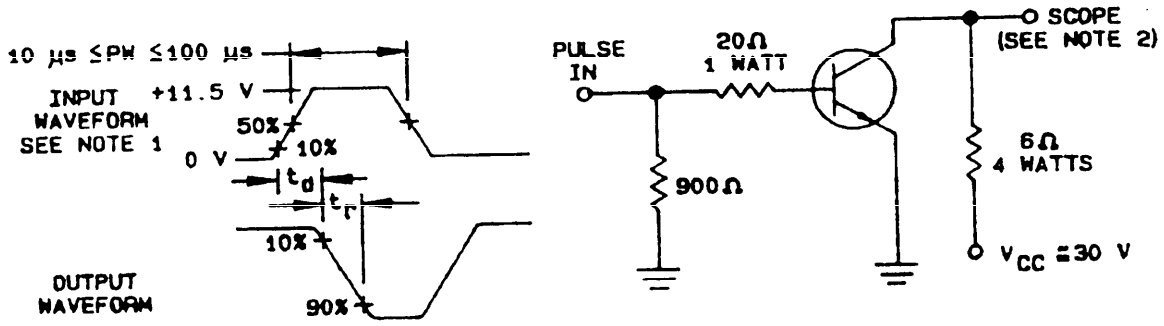
c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be within the range 25°C ≤ T_R ≤ 75°C and recorded before the test is started.

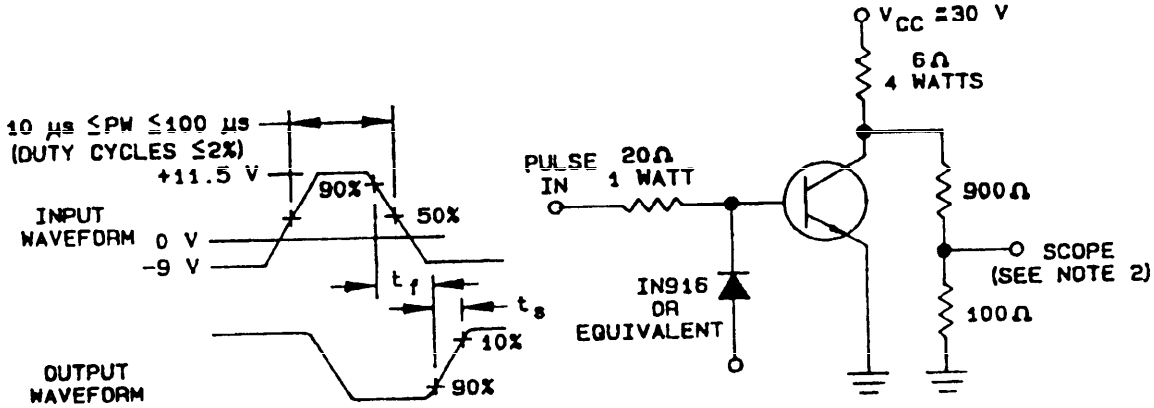
e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 1.17°C/W.

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TEST CIRCUIT A



TEST CIRCUIT B

NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20.0$ ns, $t_f \leq 1$ µs, 10 µs ≤ PW ≤ 100 µs, $Z_{OUT} = 50\Omega$, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 5$ ns, $Z_{IN} \geq 100$ kΩ, $C_{IN} \leq 12$ pF.
3. Test circuit A for t_d and t_r; test circuit B for t_s and t_f.

FIGURE 2. Pulse response test circuits.

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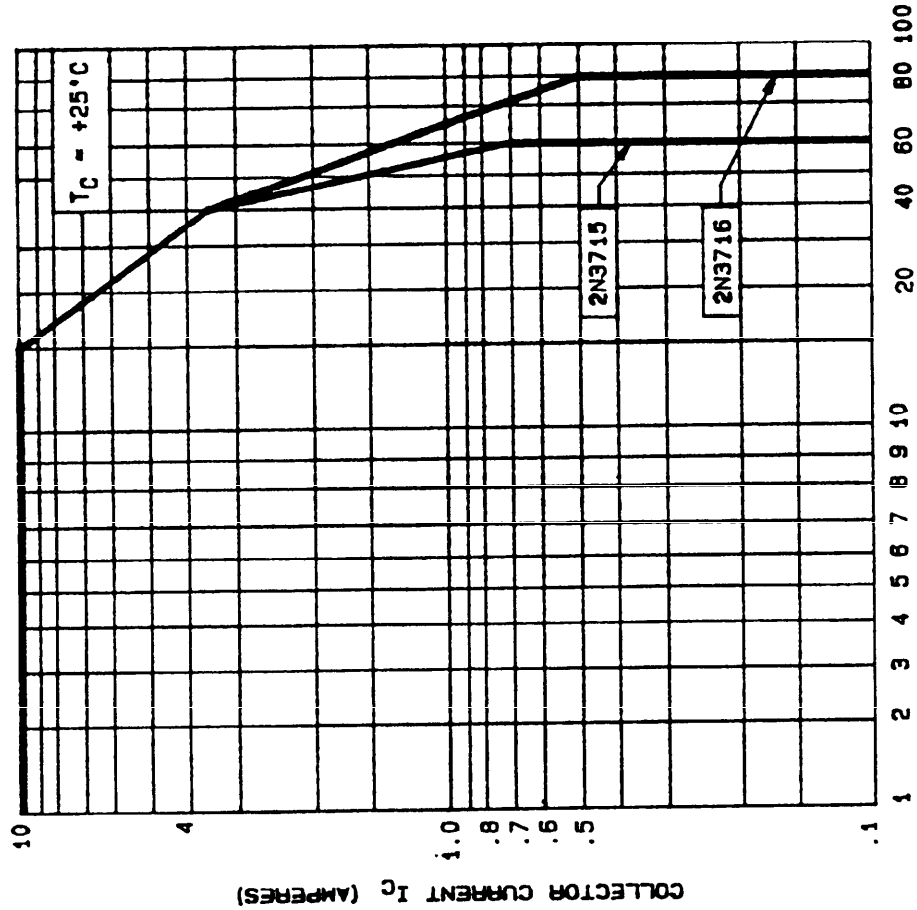


FIGURE 3. Maximum safe operating graph (dc).

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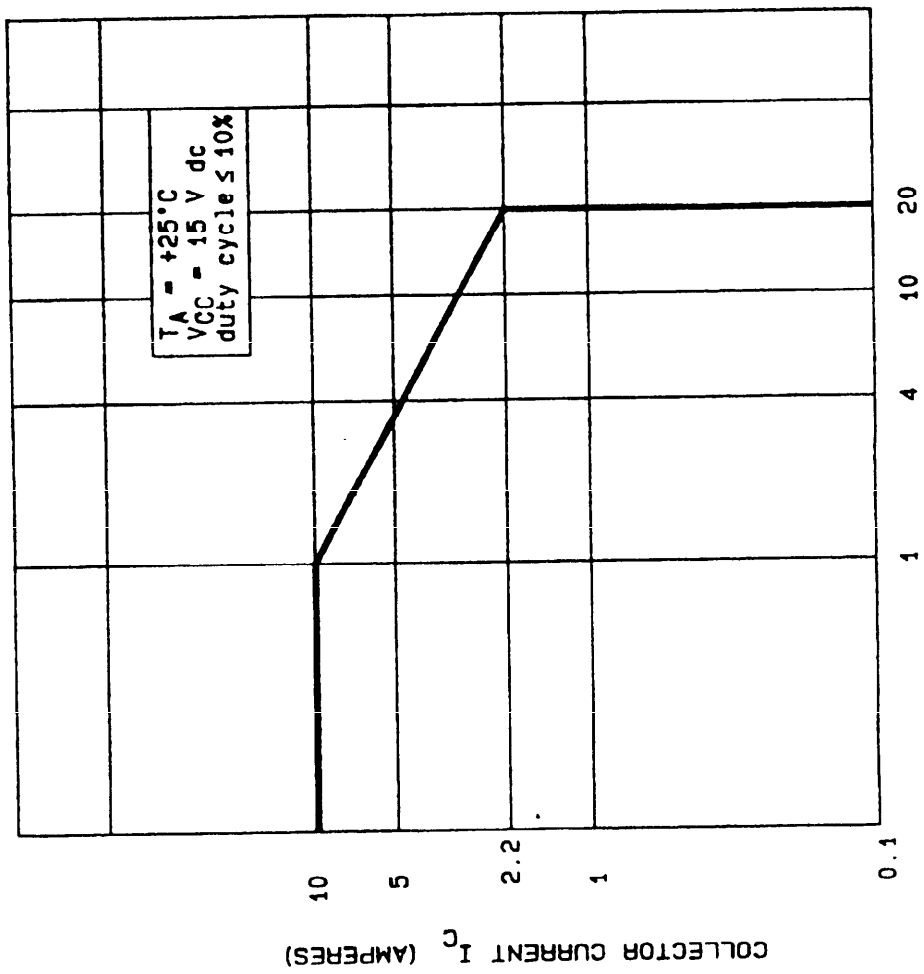


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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Check the source to verify that this is the current version before use.

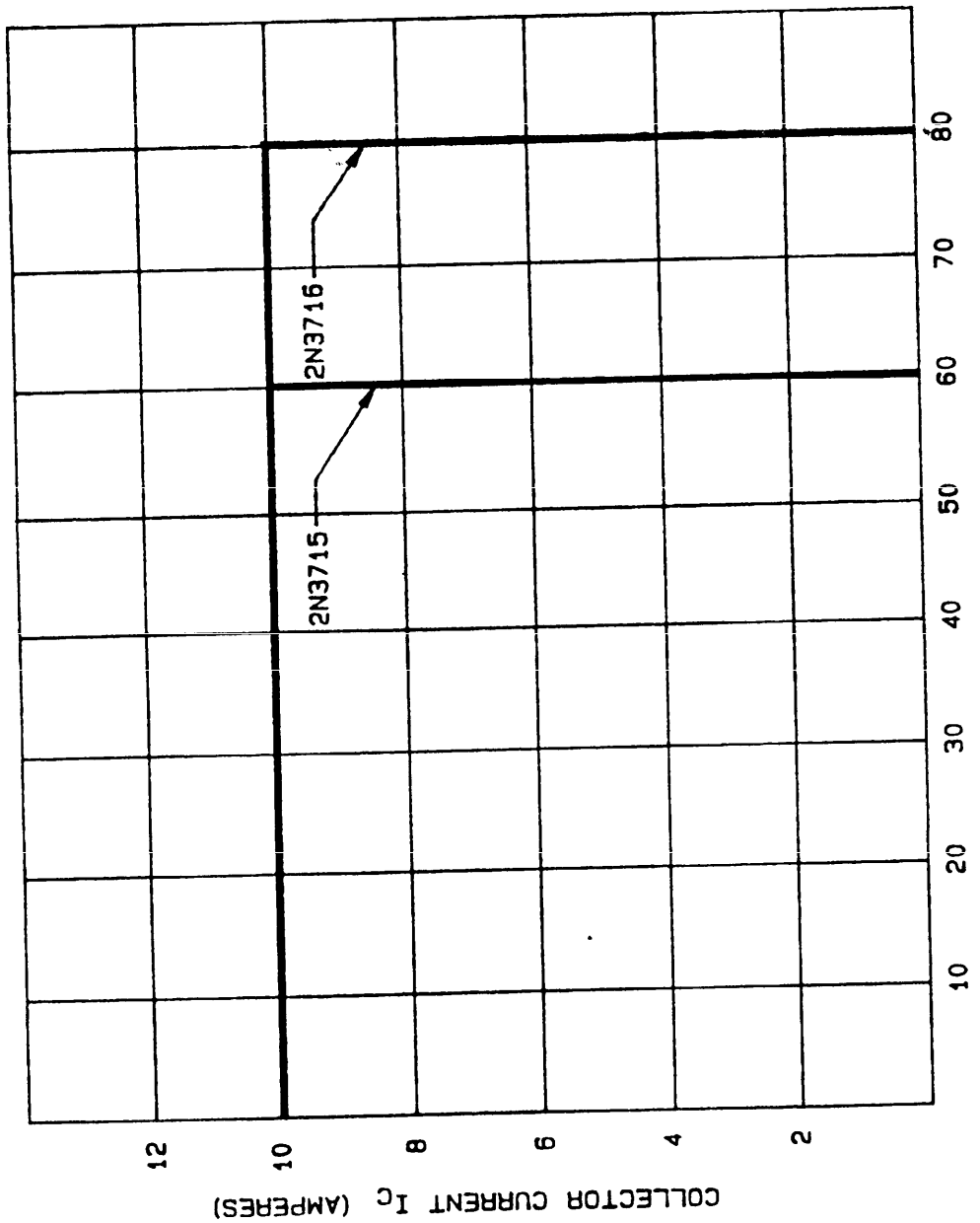
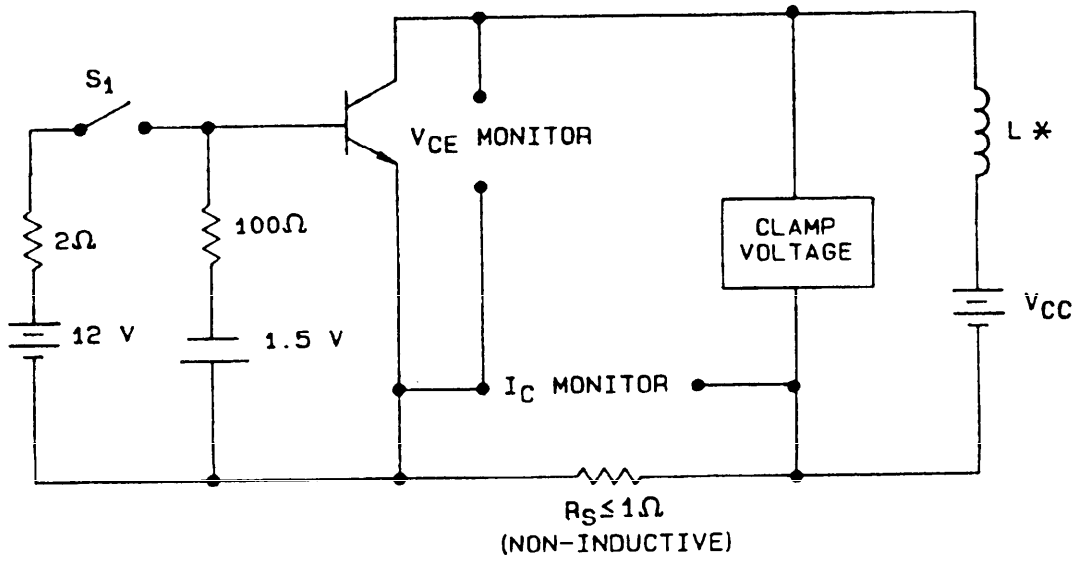


FIGURE 5. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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* L = 4 mH, .05Ω, 20 A
Q ≥ 100 at 1 kHz
(Stanford Miller CK-20 or equivalent)

Procedures:
1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails, if clamp voltage is not reached.
3. Perform specified end-point tests.

FIGURE 6. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish may be specified (see 3.3.1).

d. Product assurance level and type designator.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - AV
Navy - AS, CG, MC
Air Force - 13, 15

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1345)

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